ScopePilot

REVIEW 3 COMPLETED

Date: Oct 18, 2026

Review 3 Technical Report: Embeddings & Approved-Scope Retrieval Experiment

Student / Developer:	Vishal	Assigned Role:	AI / RAG / Change-Analysis Agent
Review Focus:	Semantic Retrieval over Approved Scope with Altered Client Wording	Working Feature Branch:	`Vishal` (Aligned with latest `origin/main`)
Automated Test Baseline:	362 / 362 TESTS PASSING ACROSS 15 SUITES	Retrieval Verification:	5 / 5 EXPERIMENTS VERIFIED DETERMINISTIC

1. Executive Summary & Review 3 Objective

In real-world software consulting engagements, clients propose change requests using colloquial, informal terminology that diverges significantly from the formal technical language of signed proposals. For example, while the proposal states `"E.164 international telephone number formatting on checkout"`, a client writes `"cellphone dial code picker on the final buying screen"`. Strict keyword queries completely fail on these requests.

Review 3 Deliverable Goals:

- **Corpus Management:** Chunk approved scope items and deliverables with strict source audit references (`sourceReference`, `proposalVersion`, `documentType`).
- **Dense Vector Embeddings:** Implement 1024-dimensional normalized vector embeddings with subword character n-grams and domain synonym clusters to bridge vocabulary gaps.
- **Evidence-Aware Semantic Search:** Calculate cosine similarity against baseline passages and categorize results into confidence bands (`strong_match`, `weak_match`, `no_evidence`).
- **Failure Analysis & Safe Fallbacks:** Document cases where candidate proximity is weak or evidence is missing, triggering safe fallbacks to `uncertain` rather than guessing.

2. Architectural Data Flow & Safeguards

The retrieval pipeline evaluates natural-language requests against baseline deliverables before passing candidates to human decision-makers:

Critical Safety Principle: A similarity score is purely a *retrieval signal* indicating semantic proximity. A high score identifies candidate evidence for review but **does not** constitute proof that the work is contractually included. Human approval remains exclusively reserved for authenticated project participants via `POST /api/change-requests/:id/decision`.

3. Approved-Scope Corpus Preparation (`corpus.service.js`)

The corpus service maintains granular scope units with immutable source citations:

Citation Ref	Title & Type	Baseline Scope Summary & Acceptance Criteria
DEL-CHK-01	Checkout Flow & Validation *(deliverable)*	Multi-step checkout flow, billing/shipping address validation, international phone formatting in E.164 standard, pre-submission input field sanitization.
DEL-CAT-01	Catalog & Faceted Search *(deliverable)*	Product catalog grid with faceted filtering by price bracket, brand, and category tags. Full-text search with keyword highlighting and sorting by date or price.
DEL-PAY-01	Stripe Payment Gateway *(deliverable)*	PCI-DSS credit/debit card processing via Stripe Elements SDK. 3D Secure 2.0 customer authentication, webhook listener, and USD/EUR settlement.
DEL-NOTIF-01	Email Notification Service *(deliverable)*	Automated transactional email dispatch via SendGrid/SMTP relay. Order receipts, tracking links, password resets. Excludes cellular SMS and app push gateways.
DEL-AUTH-01	Customer Authentication *(deliverable)*	User registration, login, and password reset workflows using stateless JWT bearer tokens and bcrypt password hashing. Stored shipping address management.
PROP-V1-SCOPE-01	Core Platform Scope Baseline *(proposal_scope_item)*	Production web-based online retail store platform running on Node.js/Express with MongoDB. Encompasses web browsing, product catalog, cart, and email receipts.

4. Canonical Retrieval Experiment Benchmark (The 5 Scenarios)

The retrieval experiment evaluates five real-world scenarios designed to test vocabulary bridging, weak candidate handling, distractor proximity, and missing evidence fallbacks:

ID	Scenario Category	Client Request Query	Top Match	Score & Status
EXP-01	**Different Wording** SEMANTIC MATCH	"Add mobile cellphone dial code picker and format check on the final buying screen"	**DEL-CHK-01** (Checkout Flow)	**69%** `candidate_found`
EXP-02	**Synonym Paraphrase** SEMANTIC MATCH	"Allow buyers to filter goods by price bracket and sort articles by arrival date"	**DEL-CAT-01** (Catalog & Search)	**94%** `candidate_found`
EXP-03	**Ambiguous Overlap** WEAK MATCH	"Send automated SMS alerts and push notifications when order status updates"	**DEL-NOTIF-01** (Email Notifications)	**50%** `weak_evidence`
EXP-04	**Out-of-Scope Distractor** PROXIMITY GUARD	"Connect decentralized web3 wallet to process crypto payments"	**DEL-PAY-01** (Stripe Gateway)	**44%** `weak_evidence`
EXP-05	**Insufficient Evidence** FALLBACK TRIGGERED	"Sync orders to internal enterprise ERP via custom warehouse AS400 connector and VPN tunnel"	**NONE** (No passage met min threshold)	**13%** `uncertain`

5. Comprehensive Failure Analysis & Safe Fallbacks

A. Resolving Vocabulary Discrepancies (EXP-01 & EXP-02)

When the client queries for "*cellphone dial code picker*" or "*filter goods by price bracket*", traditional keyword string searching yields 0% overlap with "*E.164 telephone format*" or "*product catalog faceted search*". Subword character n-grams and domain synonym clusters in `embedding.service.js` project these expressions into overlapping dense coordinates, achieving 69% and 94% similarity and correctly retrieving the relevant baseline deliverables.

B. Guarding Against False Scope Inclusions in Partial Overlaps (EXP-03)

In EXP-03, the client asks for SMS text messages and push notifications. Naive AI systems frequently observe the word "notification" and falsely declare the request in scope. Our engine scores the candidate in the weak band (50%), returning `status: "weak_evidence"` and attaching actionable clarifying questions:

```
"Top candidate DEL-NOTIF-01 matched with moderate similarity (50%), but contains significant semantic divergence from approved deliverables. Does this request extend existing deliverable DEL-NOTIF-01 or introduce a distinct technical service?"
```

C. Out-of-Scope Distractor Nearest Neighbors (EXP-04)

When evaluating cryptocurrency and Web3 requests, vector search identifies `DEL-PAY-01` (Stripe payment gateway) as the closest vector neighbor due to payment concepts. The system detects that the score falls in the weak band (44%), enforces the advisory notice that proximity is not contract approval, and prompts for a formal change-order estimate.

D. Zero-Evidence Uncertain Fallback (EXP-05)

When a request references proprietary on-premise infrastructure (AS400 warehouse connectors, VPN tunnels) with zero documentation in the approved baseline, similarity scores drop to 13% (well below the 35% minimum threshold). The engine clears candidates (`topMatch: null`), returns `status: "uncertain"`, and generates technical discovery questions for client engineering.

6. CLI Demonstration Runner Output (`npm run rag:experiment`)

The standalone CLI test runner (`backend/src/ai/scripts/runRetrievalExperiment.js`) executes all five benchmark cases with sub-5ms latency and bit-for-bit determinism:

```
================================================================================
 SUMMARY TABLE: REVIEW 3 SEMANTIC RETRIEVAL BENCHMARK
================================================================================
┌─────────┬──────────┬──────────────────────────────────────────┬──────────────┬───────┬──────────────────┬──────────┬──────────────┐
│ (index) │ expId    │ category                                 │ topRef       │ score │ status           │ deterministic │ verdict │
├─────────┼──────────┼──────────────────────────────────────────┼──────────────┼───────┼──────────────────┼──────────┼──────────────┤
│ 0       │ 'exp-01' │ 'Different Wording (Semantic Match)'     │ 'DEL-CHK-01' │ '69%' │ 'candidate_found' │ 'YES'    │ 'PASS'       │
│ 1       │ 'exp-02' │ 'Synonym Paraphrase (Catalog Search)'    │ 'DEL-CAT-01' │ '94%' │ 'candidate_found' │ 'YES'    │ 'PASS'       │
│ 2       │ 'exp-03' │ 'Ambiguous Overlap (Weak Match)'         │ 'DEL-NOTIF-01' │ '50%' │ 'weak_evidence'   │ 'YES'    │ 'PASS'       │
│ 3       │ 'exp-04' │ 'Out-of-Scope Distractor (Proximity Guard)' │ 'DEL-PAY-01' │ '44%' │ 'weak_evidence'   │ 'YES'    │ 'PASS'       │
│ 4       │ 'exp-05' │ 'Insufficient Evidence (Fallback Triggered)' │ 'NONE'       │ '13%' │ 'uncertain'       │ 'YES'    │ 'PASS'       │
└─────────┴──────────┴──────────────────────────────────────────┴──────────────┴───────┴──────────────────┴──────────┴──────────────┘
================================================================================
All 5 semantic retrieval experiments passed all criteria deterministically.
Review 3 demonstration successfully verified.
```

7. Automated Test Execution Evidence (362 / 362 Tests Passing)

The entire backend automated test suite was executed via Jest, confirming 100% test passage across all 15 test suites with zero regressions:

```
> scopepilot-backend@1.0.0 test
> jest --detectOpenHandles --runInBand

PASS tests/proposal.test.js
PASS tests/ai.retrieval.test.js
PASS tests/changeRequest.test.js
PASS tests/auth.test.js
PASS tests/milestoneDeliverable.api.test.js
PASS tests/workspace.test.js
PASS tests/project.test.js
PASS tests/integration.routes.test.js
PASS tests/ai.routes.test.js
PASS tests/validation.test.js
PASS tests/health.test.js
PASS tests/notFound.test.js
PASS tests/milestoneDeliverable.model.test.js
PASS tests/ai.mockCases.test.js
PASS tests/ai.contracts.test.js

Test Suites: 15 passed, 15 total
Tests:       362 passed, 362 total
Snapshots:   0 total
Time:        5.091 s
Ran all test suites.
```

8. HTTP REST API Routes Mounted in Express (`/api/ai/*`)

- `GET /api/ai/rag/corpus` — Returns all indexed approved scope passages with source references and metadata.
- `GET /api/ai/rag/experiments` — Returns the five canonical retrieval experiment scenarios and hypotheses.
- `GET /api/ai/rag/experiments/run` — Executes the benchmark suite over HTTP and returns aggregated performance results.
- `POST /api/ai/rag/retrieve` — Accepts dynamic queries validated via `RetrievalQueryInputSchema`, returning ranked candidate passages with similarity scores and uncertainty classifications.

9. Mentor Evaluation Questions & Prepared Answers

Mentor Question	Prepared Technical Response
What does semantic search add over keyword matching?	Semantic search retrieves candidate approved scope items even when the client uses informal phrasing, synonyms, or colloquial terminology (e.g. "cellphone dial code" → "E.164 telephone format") that would completely miss keyword queries.
Does a high similarity score prove the work is included?	No. A similarity score is purely a retrieval signal indicating lexical and conceptual proximity. The comparison engine and human decision-makers must evaluate the evidence, and the system retains uncertainty where requirements diverge.
Why keep source and version references?	Source references (`DEL-CHK-01` , `proposalVersion: 1`) guarantee auditability. When the AI agent presents candidate evidence, reviewers and project leads can verify the exact contractual document backing the analysis.
What happens when the client request has zero baseline evidence?	When similarity scores fall below the threshold (< 0.35), the engine triggers an uncertain fallback: it returns `status: "uncertain"` , clears false candidates (`topMatch: null`), and outputs structured discovery questions.

10. Next Steps: Roadmap to Review 4

1. Connect the retrieved candidate scope passages directly into the comparison engine as grounded context.
2. Interface with local Ollama LLM behind a strict structured-output schema to synthesize comparison rationales.
3. Integrate with the Change Request lifecycle (`src/modules/changeRequest/`) to auto-populate draft analysis on incoming requests.